Risk Aversion or Mistaken Beliefs?

Three Martingales and More*

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Abstract

This paper explores the confounding of risk aversion and mistaken beliefs in asset pricing. A likelihood ratio—a multiplicative martingale increment—serves as the key mathematical object that connects risk-neutral pricing, subjective belief distortions, and concerns about model misspecification. I describe three uses of likelihood ratios: (i) to encode a representative investor's risk aversion, following Lucas and Hansen; (ii) to represent mistaken beliefs of professional forecasters, following Piazzesi, Salomao, and Schneider; and (iii) to capture fears of model misspecification, following Hansen and Szőke. I show how the third use, with a tilted discounted entropy ball that accommodates concerns about particular parametric models featuring long-run risk, can generate the state-dependent uncertainty prices needed to explain hitherto puzzling term-structure movements.

*This paper is the basis for the inaugural David Backus Memorial Lecture. Dave was a pioneer in the study of affine term structure models and international risk sharing. His 2001 paper with Foresi and Telmer on affine models of currency pricing is a direct intellectual ancestor of the framework presented here. Dave insisted on connecting theory tightly to data. More important, Dave was extraordinarily generous as a colleague and teacher at NYU Stern. I thank Jonathan Payne, Jiao Shi, and Bálint Szőke and for helpful discussions that preceded my giving the Backus lecture.

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1 Introduction

Long-standing puzzles about the term structure of interest rates motivated some of Dave Backus's important work. They motvate this paper too. These puzzles are about the joint distribution of macroeconomic variables and asset returns implied by some popular benchmark macroeconomic models. The nominal yield curve usually slopes upward, the long-minus-short yield spread narrows before recessions and widens after them, and—most strikingly—long-term bond yields are nearly as volatile as short-term yields even though, under the rational-expectations expectations theory of the term structure, they should be much smoother averages of expected future short rates. This is Robert Shiller's (1979) "volatility puzzle." Sargent (1972, 1979) had tested the cross-equation restrictions on vector autoregressions implied by the rational expectations theory of the term structure, finding support for the theory's implications for conditional means but early hints of the difficulties with conditional second moments that Shiller later codified. To resolve the volatility puzzle, risk prices—or something observationally equivalent—must be large, volatile, and state-dependent. What generates those state-dependent risk prices? Risk aversion? Wrong beliefs? Fears of model misspecification? Disentangling potential sources of those puzzling risk prices is the topic this paper.

Consequences of investors' *risk aversion* and *wrong beliefs* are confounded in asset pricing data. In a rational expectations model with a risk-averse representative investor, higher mean returns compensate for higher risks. But if the representative investor holds wrong beliefs, observed average returns depend on *both* risk aversion *and* misunderstood return distributions. Wrong beliefs can contribute what from the point of view of an econometrician's baseline model look like "stochastic discount factor shocks" that have the potential to explain observed countercyclical risk prices.

This paper organizes ideas around a single mathematical device: the *likelihood ratio*, a non-negative random variable with unit mean that twists one probability distribution into another. Much of the analysis draws on two papers: Hansen, Szőke, Han, and Sargent (2020), who construct twisted probability models from tilted discounted entropy balls to price model uncertainty, and Szőke (2022), who estimates a robust control model with cross-equation restrictions implied by a representative agent's worst-case beliefs and evaluates its implications for the yield curve and survey expectations. Likelihood ratios—equivalently, multiplicative martingale increments—appear in at least four distinct roles in modern asset pricing:

Probability	**Likelihood ratio**	**Describes**
Econometric	1	macro risk factors
Risk neutral	m^{λ}_{t+1}	prices of risks
Mistaken	m^{w}_{t+1}	experts' forecasts
Doubtful	$m_{t+1} \in \mathcal{M}$	misspecification fears

Table 1: Four probability measures and associated likelihood ratios. Each likelihood ratio takes the log-normal form $m^{b}_{t+1} = \exp\big(-b_t'\varepsilon_{t+1} - \frac{1}{2}b_t'b_t\big)$ with $b_t = 0$, λ_t, w_t, or a worst-case distortion.

The "three martingales" of the sub title are the cumulative likelihood ratio processes $\{M_t\}$ formed from (i) risk prices λ_t, (ii) belief distortions w_t^*, and (iii) worst-case model distortions $\tilde{w}_t$. The "and more" refers to $\bar{w}_t$, the parametric long-run risk worry that tilts the entropy ball.

The rest of the paper proceeds as follows. Section 2 sets up the Gaussian framework in which a likelihood ratio twists probabilities while preserving a tractable structure. Section 3 reviews rational-expectations asset pricing and shows how a single likelihood ratio turns risk-neutral pricing formulas into modern asset pricing formulas. Section 4 discusses the identification challenge: the same likelihood ratio can be interpreted as risk aversion or as belief distortion. Section 5 describes the approach of Piazzesi, Salomao, and Schneider (2015), who use survey data on professional forecasters to decompose the likelihood ratio into a smaller risk price and a belief distortion. Section 7 develops the robust control theory approach, beginning with Hansen's formulation with a discounted entropy ball and culminating in Hansen and Szőke's tilted entropy ball that accommodates concerns about parametric alternatives with more long-run risk. Section 9 summarizes empirical results. Section 11 concludes with reflections on why these joint probability distributions matter for macroeconomics.

2 Risks and Beliefs

This section introduces the paper's central mathematical device: a likelihood ratio that twists one probability distribution into another. In a Gaussian setting, this twist amounts to shifting the mean of the shock distribution by a vector λ while leaving the covariance structure unchanged. The resulting framework is deliberately simple—it sets up the vocabulary and notation that will recur throughout the paper.

Let ε denote a vector of risks to be taken and priced. Suppose that, under the econometrician's probability model, ε has a standard multivariate normal density:

$$\phi(\varepsilon) \propto \exp\left(-\tfrac{1}{2}\,\varepsilon'\varepsilon\right), \qquad \varepsilon \sim \mathcal{N}(0, I).$$

Define a *likelihood ratio*

$$m(\varepsilon) = \exp\left(-\lambda'\varepsilon - \tfrac{1}{2}\,\lambda'\lambda\right) \geq 0,$$

which satisfies $E\,m(\varepsilon) = 1$ by construction. The *twisted density*

$$\hat{\phi}(\varepsilon) = m(\varepsilon)\,\phi(\varepsilon) \propto \exp\left(-\tfrac{1}{2}\,(\varepsilon+\lambda)'(\varepsilon+\lambda)\right)$$

is a $\mathcal{N}(-\lambda, I)$ density. The *relative entropy* of the twisted density with respect to the baseline density is

$$E\big[m(\varepsilon)\log m(\varepsilon)\big] = \tfrac{1}{2}\,\lambda'\lambda,$$

a convenient measure of the statistical distance between the two models.

The vector λ is the key object. Depending on context it represents risk prices, belief distortions, or worst-case mean perturbations under model uncertainty.

3 Asset Pricing with Likelihood Ratios

With the likelihood ratio in hand, we now embed it in a dynamic asset pricing framework. The key insight is that the same formulas used to price assets under risk neutrality and rational expectations carry over to settings with risk-averse or belief-distorted investors—one simply replaces the econometrician's probability measure with a twisted one. This section develops that idea in a linear Gaussian state-space setting and shows how a single likelihood ratio generates the risk-neutral dynamics that underlie modern affine term structure models.

3.1 The econometrician's model

The econometrician works with a linear Gaussian state-space system:

$$x_{t+1} = A\,x_t + C\,\varepsilon_{t+1}, \tag{1}$$

$$y_{t+1} = D\,x_t + G\,\varepsilon_{t+1}, \tag{2}$$

$$\varepsilon_{t+1} \sim \mathcal{N}(0, I),$$

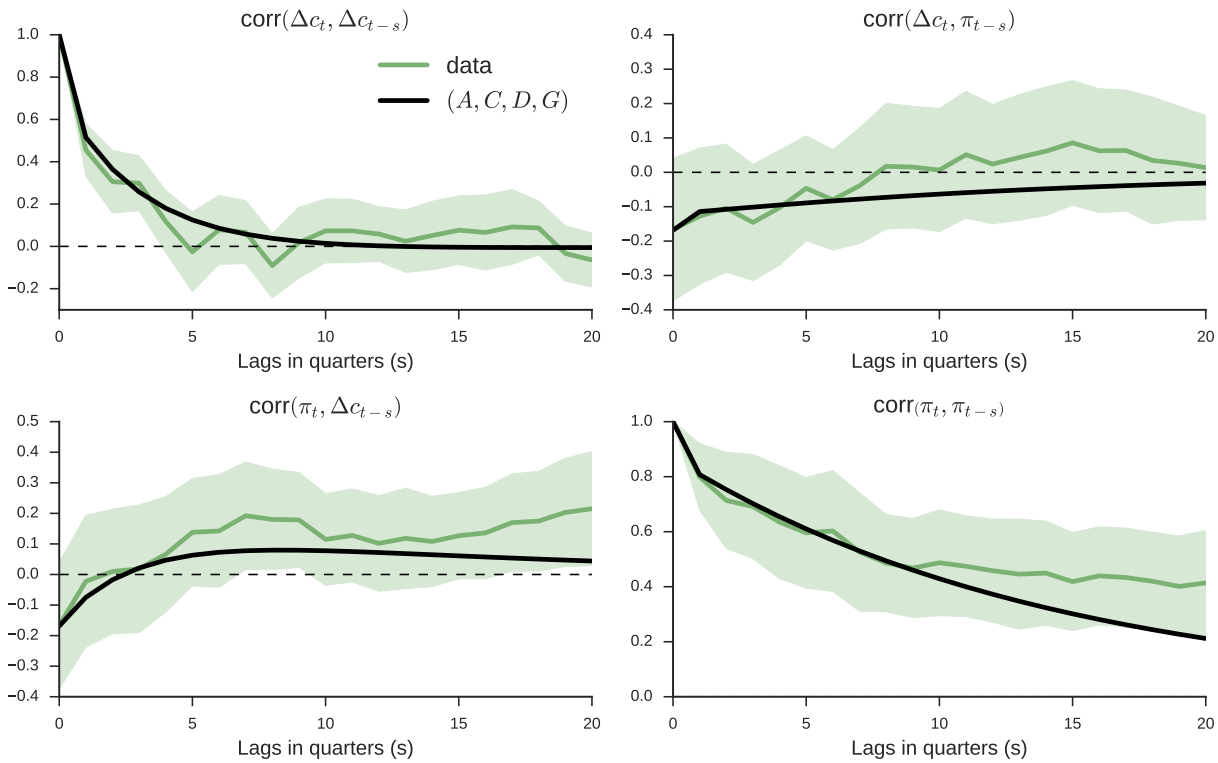

Figure 1: The econometrician's model: estimated state dynamics.

where y_{t+1} collects utility-relevant variables, $r_t = \bar{r}\, x_t$ is the risk-free one-period interest rate, and $d_t = \bar{d}\, x_t$ is the payout process from an asset. The object of interest is the price at date t of a claim to the random payout stream $\{d_{t+j}\}_{j=1}^{\infty}$.

3.2 Risk-neutral rational expectations pricing

Under rational expectations with a risk-neutral representative investor, stock prices satisfy (Shiller):

$$p_t = \exp(-r_t)\, E_t\big(p_{t+1} + d_{t+1}\big).$$

The expectations theory of the term structure of interest rates (Hicks–Shiller) prices a zero-coupon risk-free claim to one dollar at time $t + n$ as:

$$\begin{aligned} p_t(1) &= \exp(-r_t), \\ p_t(n+1) &= \exp(-r_t)\, E_t\, p_{t+1}(n), \\ p_t(n) &= \exp(B_n\, x_t). \end{aligned} \tag{3}$$

These formulas work "pretty well" for conditional means but less well for conditional variances—the Shiller "volatility puzzles."

3.3 Modern asset pricing: adding risk aversion

It would be convenient if versions of the same formulas worked even when investors are risk averse or do not have rational expectations. The likelihood ratio makes this possible.

Let

$$\begin{aligned} m^{\lambda}_{t+1} &= \exp\bigl(-\lambda_t' \, \varepsilon_{t+1} - \tfrac{1}{2}\, \lambda_t' \lambda_t\bigr)\,, \qquad \lambda_t = \lambda \, x_t, \\ E_t \, m^{\lambda}_{t+1} &= 1, \qquad m^{\lambda}_{t+1} \geq 0. \end{aligned} \tag{4}$$

The likelihood ratio m^{λ}_{t+1} distorts the conditional distribution of ε_{t+1} from $\mathcal{N}(0, I)$ to $\mathcal{N}(-\lambda \, x_t, I)$. Covariances of returns with m^{λ}_{t+1} affect mean returns—this is the channel through which risk aversion prices risks.

With this device, modern asset pricing (Lucas–Hansen for stocks, Dai–Singleton–Backus–Zin for bonds) takes the form:

$$p_t = \exp(-r_t) \, E_t\bigl(m^{\lambda}_{t+1} \, (p_{t+1} + d_{t+1})\bigr), \tag{5}$$

and for the term structure:

$$\begin{aligned} p_t(1) &= \exp(-r_t), \\ p_t(n+1) &= \exp(-r_t) \, E_t\bigl(m^{\lambda}_{t+1} \, p_{t+1}(n)\bigr), \\ p_t(n) &= \exp(B_n \, x_t). \end{aligned} \tag{6}$$

3.4 Risk-neutral dynamics

The risk-neutral representation implies twisted dynamics:

$$x_{t+1} = (A - C\lambda) \, x_t + C \, \tilde{\varepsilon}_{t+1}, \qquad \tilde{\varepsilon}_{t+1} \sim \mathcal{N}(0, I).$$

The risk-neutral dynamics assert that the shock distribution ε_{t+1} has conditional mean $-\lambda_t$ instead of 0. The dependence of $\lambda_t = \lambda \, x_t$ on the state modifies the dynamics relative to the econometrician's model.

3.5 Expectation under a twisted distribution

The mathematical expectation of y_{t+1} under the probability distribution twisted by likelihood ratio m_{t+1} is

$$\tilde{E}_t\, y_{t+1} = E_t\, m_{t+1}\, y_{t+1}.$$

With respect to the risk-neutral dynamics, the term structure theory becomes:

$$\begin{aligned} p_t(1) &= \exp(-r_t), \\ p_t(n+1) &= \exp(-r_t)\, \tilde{E}_t\, p_{t+1}(n), \\ p_t(n) &= \exp(\tilde{B}_n\, x_t). \end{aligned}$$

These are the same formulas as rational-expectations asset pricing, but mathematical expectations are taken with respect to a probability measure twisted by risk aversion.

4 The Identification Challenge

The elegance of the likelihood ratio device comes at a cost: the same mathematical object can represent very different economic forces. This section makes the observational equivalence precise. Because asset pricing formulas depend on λ_t but not on its interpretation, the data alone cannot tell us whether observed risk premia reflect the representative investor's risk aversion or systematically mistaken beliefs. Resolving this issue requires either additional data or additional theory—the subjects of the sections that follow.

The vector λ_t can be interpreted as either:

- a risk price vector expressing the representative agent's risk aversion, or
- the representative agent's belief distortion relative to the econometrician's model.

The asset pricing formulas (5)–(6) are identical under both interpretations, and so are the econometric fits. Relative to the model of a risk-averse representative investor with rational expectations, a model of a risk-neutral investor with appropriately mistaken beliefs produces observationally equivalent predictions. This insight was articulated by Hansen, Sargent, and Tallarini (1999) and Piazzesi, Salomao, and Schneider (2015).

To distinguish risk aversion from belief distortion, one needs either more information (the PSS approach using survey data) or more theory (the Hansen and Szőke robust control approach), or both (Szőke).

5 More Information: Experts' Forecasts

One natural strategy for breaking the identification deadlock is to bring in additional data. Piazzesi, Salomao, and Schneider (2015) pursue exactly this approach by exploiting surveys of professional forecasters. If forecasters' reported expectations differ systematically from those implied by the econometrician's model, the discrepancy can be attributed to belief distortions, leaving a smaller residual to be explained by risk aversion alone. This section describes their framework and its empirical findings.

5.1 The PSS framework

Piazzesi, Salomao, and Schneider (2015, henceforth PSS) exploit data on professional forecasters' expectations to decompose the likelihood ratio into risk prices and belief distortions. Their setup posits:

- The representative agent's risk aversion leads him to price risks ε_{t+1} with prices $\lambda_t^* = \lambda^* x_t$.
- The representative agent has twisted beliefs $(A^*, C) = (A - Cw^*, C)$ relative to the econometrician's model (A, C).
- Professional forecasters use the twisted beliefs (A^*, C) to answer survey questions about their forecasts.

The question PSS assume experts think they are answering is: "What forecasts underlie your investment advice decisions?"

5.2 Estimation strategy

PSS proceed as follows:

1. Use data $\{x_t\}_{t=0}^T$ to estimate the econometrician's model A, C.
2. Project experts' forecasts $\{\hat{x}_{t+1}\}$ on x_t to obtain $\hat{x}_{t+1} = A^* x_t$ and interpret A^* as a belief distortion.
3. Back out the mean distortion $w^* x_t = -C^{-1}(A^* - A)\, x_t$ to the density of ε_{t+1}.
4. Reinterpret the λ estimated by the rational-expectations econometrician as $\lambda = \lambda^* + w^*$, where $\lambda_t^* = \lambda^* x_t$ is the (smaller) price of risk vector actually charged by the representative agent with distorted beliefs.

An econometrician who mistakenly imposes rational expectations estimates risk prices λ_t that sum two parts:

- smaller risk prices λ_t^* actually charged by the erroneous-beliefs representative agent, and
- conditional mean distortions w_t^* of the risks ε_{t+1} that the twisted-beliefs representative agent's model displays relative to the econometrician's.

5.3 Empirical specification

The key variables in PSS's state-space system are:

- the level and slope of the yield curve (short rate; spread between 5-year and short rate),
- inflation, and
- conditional expectations of these variables.

5.4 PSS successes

PSS find $w^* \neq 0$:

- The expert's model differs systematically from the econometrician's.
- Experts perceive the level and slope of the yield curve to be more persistent than the econometrician's estimates imply.
- Subjective risk prices $\lambda^* x_t$ vary less than the λx_t estimated by the rational-expectations econometrician.

5.5 Why are beliefs distorted?

PSS offer no explanation for why beliefs are distorted. Are they mistakes? Ignorance of good econometrics? And how distorted are they in terms of the divergence of probabilities? These questions motivate a theory of belief distortions rooted in robust control theory.

An intriguing interpretive question lurks here: are professional forecasters reporting their best guess of the future, or the scenario that rationalizes their portfolio decisions? If the latter, their "mistakes" may not be mistakes at all but instead reflect a pessimistic model that they use precisely because they fear misspecification. Section 7 develops a theory in

which this distinction is formalized: the robust control agent's worst-case model generates forecasts that look like the experts' distorted beliefs, but arise from an optimizing response to model uncertainty rather than from ignorance.

6 Rational Expectations and Its Alternatives

Before developing a theory of belief distortions, it is worth pausing to ask why a rational expectations model might not be a good approximation.[1] The standard justification treats rational expectations as the outcome of learning from a long history—but how long is long enough? This section reviews the conditions under which learning converges to rational expectations and highlights circumstances, particularly those involving persistent dynamics, in which convergence is so slow that model uncertainty becomes economically first-order.

6.1 The 1970s–1980s informal justification

The standard justification for rational expectations treats it as the outcome of learning from an infinite history: least-squares learning converges to rational expectations. The argument requires that:

- agents know correct functional forms, and
- a stochastic approximation argument partitions dynamics into fast parts (justifying a law of large numbers) and slow parts (justifying an ODE).

However, long intertemporal dependencies make rates of convergence slow. For example, distinguishing two Gaussian AR(1) processes whose persistence parameters differ by 0.01 (say $\rho = 0.98$ versus $\rho = 0.99$) at the 5% level requires several hundred years of quarterly data. For the long-run risk specifications considered by Bansal and Yaron (2004), the required sample lengths can exceed a millennium. This is not a failure of econometrics—it is the statistical reality that makes model uncertainty economically important.

6.2 Good econometricians

Good econometricians have limited data and only hunches about functional forms. They fear that their fitted models are incorrect. An agent who is like a good econometrician:

[1] Sargent (2008) discusses the intellectual history of rational expectations and adaptive learning, framing the tension between them as one between "intelligent design" and "evolution." The introduction to Lucas and Sargent (1981) lays out the case for rational expectations as a discipline on econometric practice.

- has a parametric model estimated from limited data,
- acknowledges that many other specifications fit nearly as well—other parameter values, other functional forms, omitted variables, neglected nonlinearities, and history dependencies,
- fears that one of those other models actually prevails, and
- seeks "good enough" decisions under *all* such alternative models—*robustness*.

7 A Theory of Belief Distortions: Robust Control

Having seen that survey data point to systematic belief distortions and that statistical theory explains why such distortions are hard to eliminate through learning, we now develop a theory that rationalizes them. Robust control theory provides a framework in which a decision maker who acknowledges that her model may be misspecified optimally distorts her probability assessments toward a worst-case scenario. The resulting worst-case beliefs look like the "mistakes" identified by PSS, but arise from a coherent response to model uncertainty rather than from ignorance.

7.1 Hansen's dubious agent

Inspired by robust control theory, consider a dubious investor who:

- shares the econometrician's model A, C, D, G,
- expresses doubts by using a continuum of likelihood ratios to form a *discounted entropy ball* of size η around the econometrician's model,
- wants a valuation that is good for every model in the entropy ball, and
- constructs a lower bound on values and a worst-case model that attains it.

7.1.1 Why discounted entropy?

Discounted entropy includes models that undiscounted entropy excludes. Undiscounted entropy over infinite sequences excludes many models that are very difficult to distinguish from the econometrician's model with limited data, because undiscounted entropy includes only models that share laws of large numbers. To see why this matters, note that undiscounted entropy effectively requires alternative models to share the same long-run averages as the

baseline—it excludes models that differ only in persistent, low-frequency dynamics. But those are precisely the models that are hardest to distinguish with finite data and that matter most for long-lived asset prices. Discounted entropy, by treating future divergences less severely, admits these statistically elusive but economically important alternatives into the set of models that the dubious agent contemplates.

7.2 Valuation under the econometrician's model

Taking the log consumption process to be linear Gaussian with shocks $\varepsilon_{t+1} \sim \mathcal{N}(0, I)$:

$$c_{t+1} - c_t = D\, x_t + G\, \varepsilon_{t+1}, \tag{7}$$

$$x_{t+1} = A\, x_t + C\, \varepsilon_{t+1}, \tag{8}$$

the value function is

$$V(x_0, c_0) := E\left[\sum_{t=0}^{\infty} \beta^t\, c_t \;\middle|\; x_0, c_0\right] = c_0 + \beta\, E[V(x_1, c_1) \mid x_0, c_0]\,.$$

7.3 The sequence problem

The dubious agent solves:

$$\begin{aligned} J(x_0, c_0 \mid \eta) := \min_{\{m_{t+1}\}_{t=0}^{\infty}} \; & E\left[\sum_{t=0}^{\infty} \beta^t\, M_t\, c_t \;\middle|\; x_0, c_0\right] \\ \text{s.t.} \quad & c_{t+1} - c_t = D\, x_t + G\, \varepsilon_{t+1}, \\ & x_{t+1} = A\, x_t + C\, \varepsilon_{t+1}, \\ & E\left[\sum_{t=0}^{\infty} \beta^t\, M_t\, E[m_{t+1} \log m_{t+1} \mid x_t, c_t] \;\middle|\; x_0, c_0\right] \le \eta, \\ & M_{t+1} = M_t\, m_{t+1}, \quad E[m_{t+1} \mid x_t, c_t] = 1, \quad M_0 = 1. \end{aligned} \tag{9}$$

The likelihood ratio process $\{M_t\}_{t=0}^{\infty}$ is a multiplicative martingale.

7.4 Entropy and the likelihood ratio

With the log-normal likelihood ratio

$$m_{t+1} := \exp\left(-\tfrac{w_t' w_t}{2} - w_t'\, \varepsilon_{t+1}\right),$$

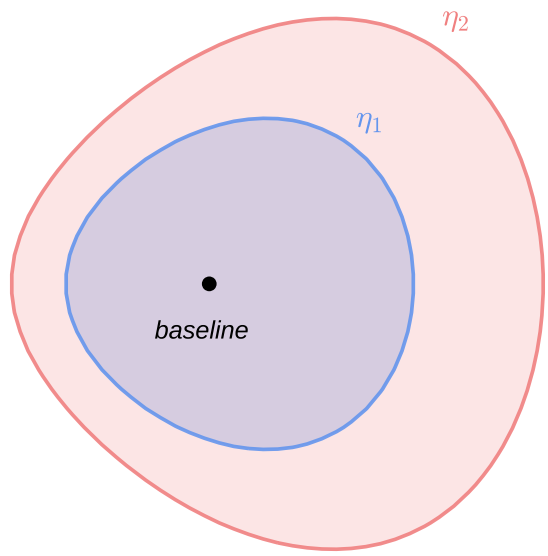

Figure 2: Discounted entropy ball around the econometrician's model.

conditional entropy takes the simple form

$$E[m_{t+1} \log m_{t+1} \mid x_t, c_t] = \tfrac{1}{2}\, w_t' w_t.$$

Substituting this into (9) yields the reformulated problem:

$$\begin{aligned} J(x_0, c_0 \mid \eta) := \min_{\{w_t\}_{t\geq 1}} \; & E^w \left[\sum_{t=0}^{\infty} \beta^t \, c_t \;\middle|\; x_0, c_0 \right] \\ \text{s.t.} \quad & c_{t+1} - c_t = D\, x_t + G\,(\tilde{\varepsilon}_{t+1} - w_t), \qquad \tilde{\varepsilon}_{t+1} \sim \mathcal{N}(0, I), \\ & x_{t+1} = A\, x_t + C\,(\tilde{\varepsilon}_{t+1} - w_t), \\ & \tfrac{1}{2}\, E^w \left[\sum_{t=0}^{\infty} \beta^t \, w_t' w_t \;\middle|\; x_0, c_0 \right] \leq \eta. \end{aligned} \tag{10}$$

7.5 Outcome of the Hansen model

The worst-case mean distortion turns out to be a constant vector:

$$w_t = \bar{w}.$$

The consequence is that the contribution of w_t to risk prices is state-independent. This does *not* help explain countercyclical prices of risk (or prices of model uncertainty).

8 Tilting the Entropy Ball

The basic robust control model of the previous section has a limitation: it generates worst-case distortions that are constant across states, and therefore cannot explain the countercyclical variation in risk prices that the data demand. This section shows how Hansen and Szőke overcome this limitation by *tilting* the entropy ball—requiring it to include particular parametric alternatives, such as models with more long-run risk. The tilt makes the entropy constraint bind differently across states, producing the state-dependent uncertainty prices needed to match term-structure dynamics.

8.1 Hansen and Szőke's more refined dubious agent

To generate state-dependent uncertainty prices, Hansen and Szőke introduce a more refined dubious agent who:

- shares the econometrician's model A, C, D, G,
- expresses doubts by using a continuum of likelihood ratios to form a discounted entropy ball around the econometrician's model,
- also insists that some martingales representing particular alternative *parametric* models be included in the discounted entropy ball.

The inclusion of those alternative parametric models *tilts* the entropy ball, which affects the worst-case model in a way that can produce countercyclical uncertainty prices.

8.2 Concern about other parametric models

The investor wants to include particular alternative models with

$$E_t[\bar{m}_{t+1} \log \bar{m}_{t+1}] = \tfrac{1}{2}\, \bar{w}_t' \, \bar{w}_t = \xi(x_t)$$

and discounted entropy

$$E^{\bar{w}} \left[\sum_{t=0}^{\infty} \beta^t \, \xi(x_t) \;\middle|\; x_0, c_0 \right].$$

This is accomplished by replacing the earlier entropy constraint with

$$\tfrac{1}{2}\, E^w \left[\sum_{t=0}^{\infty} \beta^t \, w_t' w_t \;\middle|\; x_0, c_0 \right] \leq E^w \left[\sum_{t=0}^{\infty} \beta^t \, \xi(x_t) \;\middle|\; x_0, c_0 \right]. \tag{11}$$

8.3 Yaron's bazooka

Bansal and Yaron (2004) mix long-run risk (LRR) with Epstein–Zin (EZ) preferences. LRR is statistically difficult to detect and estimate, but an Epstein–Zin agent or a dubious agent really hates LRR. Under certain conditions, the EZ value function is the indirect utility function of a dubious agent. This sets the stage for big risk prices—but not countercyclical ones.

Remark 1 (EZ–robust control equivalence)**.** The equivalence between Epstein–Zin preferences and robust control, established by Hansen and Sargent (2001) and Maenhout (2004), implies that EZ preferences with unit elasticity of intertemporal substitution and risk aversion $\gamma > 1$ are observationally equivalent to expected utility with $\gamma = 1$ combined with a penalty parameter $\theta = 1/(\gamma - 1)$ governing robustness. The higher the risk aversion under EZ, the smaller the entropy ball the robust agent is willing to tolerate—she fears a narrower but more intense set of model misspecifications.

8.4 Concern about bigger long-run risk in the Szőke model

Inspired by Bansal and Yaron (2004), an agent fears particular long-run risks expressed by

$$x_{t+1} = \bar{A}\, x_t + C\, \tilde{\varepsilon}_{t+1}.$$

This corresponds to $\bar{w}_t = \bar{w}\, x_t$ with

$$\bar{w} = -C^{-1}(\bar{A} - A),$$

which implies a quadratic ξ function:

$$\xi(x_t) := x_t' \, \bar{w}' \, \bar{w} \, x_t =: x_t' \, \Xi \, x_t.$$

8.5 State-dependent contributions to the entropy constraint

The time-t contributions to the right-hand side of (11) relax the discounted entropy constraint in states x_t in which $\xi(x_t)$ is larger. This sets the stage for state-dependent mean

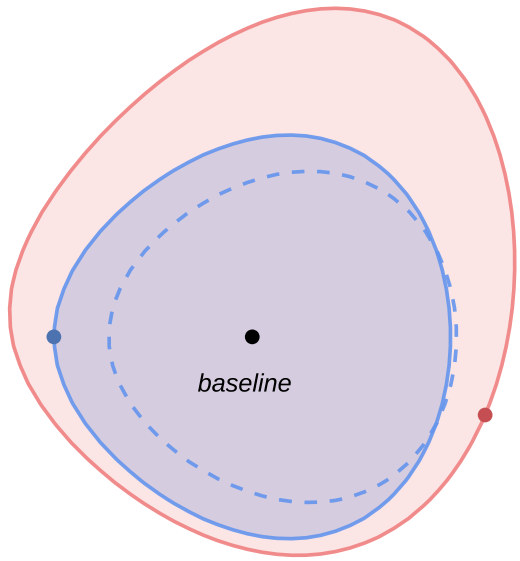

Figure 3: Tilted discounted entropy balls. Including particular parametric alternatives with more long-run risk tilts the entropy ball and generates state-dependent worst-case distortions.

distortions in the worst-case model, which in turn ignite countercyclical market prices of uncertainty.

8.6 The Szőke agent's sequence problem

The resulting linear-quadratic problem is:

$$J(x_0, c_0 \mid \Xi) := \max_{\tilde{\theta} \geq 0} \min_{\{w_t\}_{t \geq 1}} E^w \left[\sum_{t=0}^{\infty} \beta^t c_t + \tilde{\theta} \, \tfrac{1}{2} \sum_{t=0}^{\infty} \beta^t \big(w_t' w_t - x_t' \, \Xi \, x_t \big) \;\middle|\; x_0, c_0 \right] \tag{12}$$

$$\text{s.t.} \quad c_{t+1} - c_t = D \, x_t + G \, (\tilde{\varepsilon}_{t+1} - w_t), \qquad \tilde{\varepsilon}_{t+1} \sim \mathcal{N}(0, I),$$

$$x_{t+1} = A \, x_t + C \, (\tilde{\varepsilon}_{t+1} - w_t).$$

The worst-case shock mean distortion is

$$\tilde{w}_t = \tilde{w} \, x_t,$$

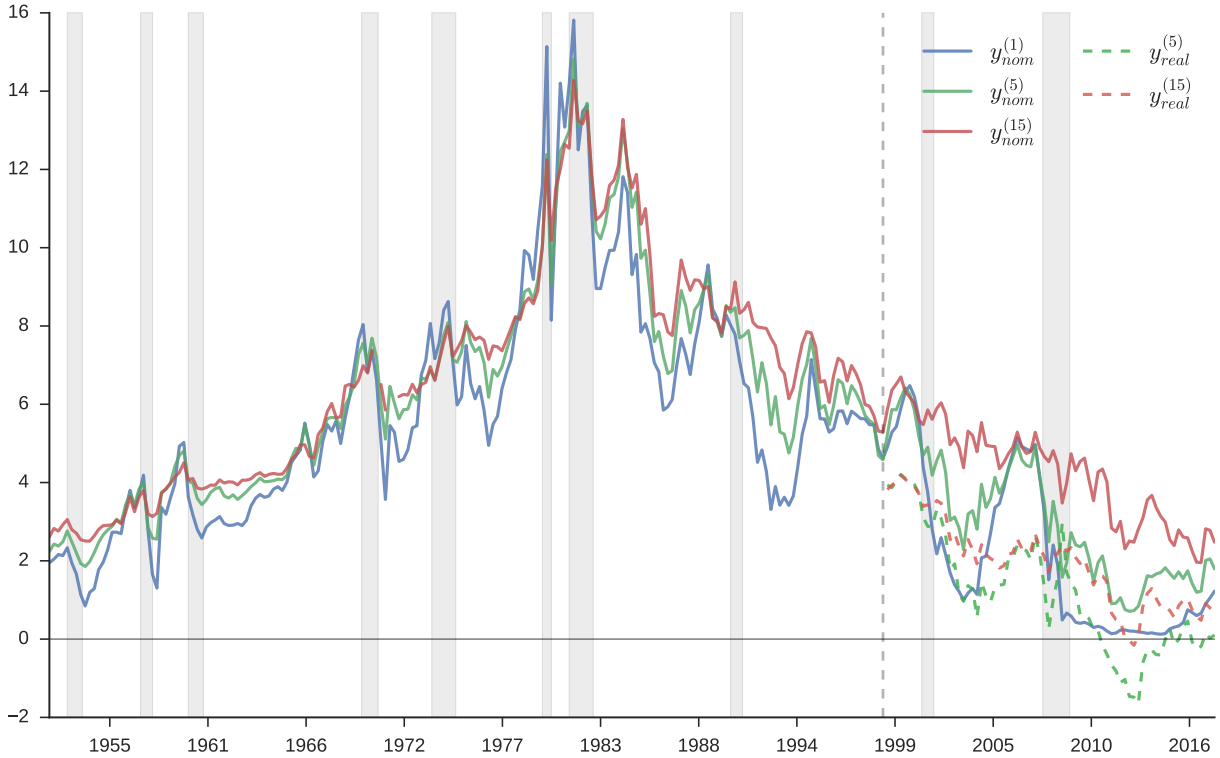

Figure 4: U.S. term structure of interest rates.

and the worst-case model is $(\tilde{A}, C, \tilde{D}, G)$ with

$$\tilde{A} = A - C\,\tilde{w},$$
$$\tilde{D} = D - G\,\tilde{w}.$$

9 Empirical Results

We now confront the theoretical framework with data. This section reviews the key empirical patterns in the U.S. term structure of interest rates—the upward-sloping yield curve, the predictive power of the yield spread for recessions, and Shiller's volatility puzzle—and assesses how well different models account for them. The centerpiece is Szőke's (2022) estimation of the tilted entropy ball model, which imposes cross-equation restrictions linking the worst-case model to both bond prices and survey forecasts.

9.1 The U.S. term structure

Several recognized patterns characterize the U.S. term structure:

- The nominal yield curve usually slopes upward.
- The long-minus-short yield spread narrows before U.S. recessions and widens after them.
- Consequently, the slope of the yield curve helps predict aggregate inputs and outputs.
- Long and short yields are almost equally volatile (the "Shiller puzzle").
- To solve the Shiller puzzle, risk prices (or something observationally equivalent) must depend on volatile state variables.

Specific historical episodes bring these patterns to life. The yield curve inverted before each of the last seven U.S. recessions. In the Szőke model, inversions correspond to states where $\xi(x_t)$ is small—the agent's long-run risk worry is muted—which tightens the entropy constraint and compresses risk prices. Conversely, during the 2008–09 financial crisis and the March 2020 COVID shock, term premia spiked as uncertainty about long-run consumption growth surged, exactly the pattern that the tilted entropy ball is designed to capture.

9.2 Challenges and responses

Table 2 summarizes how various models perform against these empirical challenges.

	Average slope	Slopes near recessions	Volatile long yield
Lucas (1978)	no	no	no
Epstein–Zin with LRR (PS (2007), HS (2001))	maybe	yes	no
PSS (2015)	built-in	built-in	yes
Szőke (2022)	YES	yes	yes

Table 2: Challenges and model performances. Affine risk prices with persistent consumption growth can address the second column; the third column requires state-dependent prices of risk.

9.3 Szőke's empirical strategy

Szőke proceeds in two stages:

Stage I: Estimation.

1. Use $\{x_t, c_t\}_{t=0}^T$ to estimate the econometrician's A, C, D, G.

2. View Ξ as a matrix of additional free parameters and estimate them simultaneously with risk prices $\tilde{\lambda}\, x_t$ from data $\{p_t(n+1)\}_{n=1}^N$, $t = 0, \ldots, T$, by imposing cross-equation restrictions:

$$
\begin{aligned}
p_t(n+1) &= \exp(-r_t)\, E_t\Big[m_{t+1}^{\tilde{w}}\, m_{t+1}^{\tilde{\lambda}}\, p_{t+1}(n)\Big], \\
m_{t+1}^{\tilde{w}} &= \exp\Big(-\tilde{w}_t'\, \varepsilon_{t+1} - \tfrac{\tilde{w}_t'\, \tilde{w}_t}{2}\Big), \\
m_{t+1}^{\tilde{\lambda}} &= \exp\Big(-\tilde{\lambda}_t'\, \varepsilon_{t+1} - \tfrac{\tilde{\lambda}_t'\, \tilde{\lambda}_t}{2}\Big),
\end{aligned}
\tag{13}
$$

   where E_t is taken with respect to the econometrician's model and $\tilde{w}_t = \tilde{w}\, x_t$ is the dubious investor's worst-case model.

Stage II: Assessment.

1. Assess improvements in predicted behavior of the term structure.

2. Use estimated worst-case dynamics to form distorted forecasts $\tilde{x}_{t+1} = (A - C\,\tilde{w})\, x_t$ and compare them to those of professional forecasters.

3. Compute discounted relative entropy of the worst-case twisted model $(A - C\,\tilde{w}, C, D - G\,\tilde{w}, G)$ relative to the econometrician's model (A, C, D, G), and use it together with Chernoff–Newman–Stuck entropy measures to assess the difficulty of distinguishing the two models.

9.4 Interpretations of conditional mean distortions

The conditional mean distortions $w\, x_t$ admit distinct interpretations:

- In PSS, w_t^* is a vector of "mistakes" or "suboptimal forecasts."

- In Szőke, $\tilde{w}_t$ is a vector of "model uncertainty" prices: compensations the representative agent charges to bear risks ε_{t+1} with unknown probability distribution.

9.5 Insights from Szőke's model

Szőke's framework delivers:

1. A theory of state-dependent belief distortions $\tilde{w}_t = \tilde{w}\, x_t$.

2. A theory about the question that professional forecasters answer: they respond with their worst-case model because they hear "tell me forecasts that rationalize your (max-min) decisions."

3. A way to measure the size of belief distortions relative to the econometrician's model.

He uses his estimated Ξ matrix:

- to infer an equivalence class of alternative parametric models parameterized by $\bar{w}$ that concern the representative investor—these models have more long-run risk than the econometrician's model;

- to infer a worst-case mean distortion $\tilde{w}\, x_t$ whose state dependence causes the term structure to move with x_t in ways that explain hitherto puzzling term-structure movements, including Shiller's "volatility puzzle."

9.6 Three probability twisters

To summarize, three distinct probability twisters play roles in this analysis:

- w_t^*: Piazzesi, Salomao, and Schneider's mistaken agent;

- $\bar{w}_t$: Szőke's especial LRR parametric worry;

- $\tilde{w}_t$: Szőke's worst-case model.

10 Cross-Equation Restrictions and Estimation

A recurring theme throughout the rational expectations econometrics literature is that equilibrium models impose testable restrictions across the equations of a vector autoregression.[2] This section highlights how the robust control framework preserves that desirable feature: even though the decision maker's beliefs depart from the econometrician's model, the departure is disciplined by the worst-case optimization problem, yielding a new but equally sharp set of cross-equation restrictions that can be taken to the data.

A key appeal of the robust control approach is that it lets us deviate from rational expectations while still preserving a set of powerful cross-equation restrictions on decision makers' beliefs. Hansen and Sargent (1991) provided systematic methods for deriving and imposing the cross-equation restrictions on vector autoregressions that are implied by exact

[2] See Lucas and Sargent (1981)

linear rational expectations models; those methods set the stage for the estimation strategy used here. As Szőke puts it:

> An appealing feature of robust control theory is that it lets us deviate from rational expectations, but still preserves a set of powerful cross-equation restrictions on decision makers' beliefs… Consequently, estimation can proceed essentially as with rational expectations econometrics. The main difference is that now restrictions through which we interpret the data emanate from the decision maker's best response to a worst-case model instead of to the econometrician's model.

11 Who Cares?

Joint probability distributions of interest rates and macroeconomic shocks are important throughout macroeconomics:

- **Costs of aggregate fluctuations.** Welfare assessments of business cycles depend sensitively on how risks are priced. Lucas (1987) calculated the welfare cost of business cycles at roughly 0.05% of lifetime consumption under standard risk aversion. But if the representative agent fears model misspecification in the sense formalized above, that cost can rise by one or two orders of magnitude (Hansen, Sargent, and Tallarini 1999), because the agent evaluates fluctuations under a pessimistic probability model that amplifies downside risk.

- **Consumption Euler equations.** The "New Keynesian IS curve" is a log-linearized consumption Euler equation whose risk adjustments are controlled by the stochastic discount factor.

- **Optimal taxation and government debt management.** Government bond prices embed risk prices whose state dependence matters for optimal fiscal policy.

- **Central bank expectations management.** Forward guidance works by shifting the term structure, an exercise whose effects depend on the same likelihood ratios studied here.

- **Long-run risk and secular stagnation.** The Bansal–Yaron long-run risk hypothesis is difficult to detect statistically, yet an agent who fears it in the sense formalized above may behave very differently than one who does not.

Understanding whether observed asset prices reflect risk aversion, mistaken beliefs, or fears of model misspecification—and quantifying each component—is therefore essential for both positive and normative macroeconomics.

A Multiplier Preferences

The multiplier preference version of the dubious agent's problem is:

$$W(x_0, c_0 \mid \theta) := \min_{\{m_{t+1}\}_{t\geq 0}} E\left[\sum_{t=0}^{\infty} \beta^t M_t\big(c_t + \theta\, m_{t+1} \log m_{t+1}\big) \,\middle|\, x_0, c_0\right], \tag{14}$$
$$M_{t+1} = M_t\, m_{t+1}, \quad E[m_{t+1} \mid x_t, c_t] = 1, \quad M_0 = 1.$$

The recursive formulation is:

$$\begin{aligned} W(x_t, c_t \mid \theta) &= c_t + \min_{m_{t+1}} E\big[m_{t+1}\big[\beta W(x_{t+1}, c_{t+1}) + \theta \log m_{t+1}\big] \mid x_t, c_t\big] \\ &= c_t - \theta \log E\left[\exp\left(-\frac{\beta W(x_{t+1}, c_{t+1})}{\theta}\right) \,\middle|\, x_t, c_t\right] \\ &=: c_t + T_t[\beta W(x_{t+1}, c_{t+1})]\,, \end{aligned} \tag{15}$$

where the right-hand side is attained by

$$m^*_{t+1} \propto \exp\left(-\frac{W(x_{t+1}, c_{t+1})}{\theta}\right).$$

Relationship between multiplier and constraint problems. By Lagrange multiplier theory,

$$W(x_t, c_t \mid \tilde{\theta}) = J(x_t, c_t \mid \eta) + \tilde{\theta}\, \eta.$$

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